

INTERNATIONAL STANDARD

**Low-voltage electrical installations –
Part 7-708: Requirements for special installations or locations – Caravan parks,
camping parks and similar locations**



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INTERNATIONAL STANDARD

**Low-voltage electrical installations –
Part 7-708: Requirements for special installations or locations – Caravan parks,
camping parks and similar locations**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

LOW-VOLTAGE ELECTRICAL INSTALLATIONS –

**Part 7-708: Requirements for special installations or locations –
Caravan parks, camping parks and similar locations**

FOREWORD

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International Standard IEC 60364-7-708 has been prepared by IEC technical committee 64: Electrical installations and protection against electric shock.

This third edition cancels and replaces the second edition published in 2007. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- requirements to protect against impact has changed: equipment installed in a campsite now has to be protected against impact of high severity AG3 (minimum degree of protection IK 08);
- socket outlets and connectors are now required to be interlocked in accordance with IEC standards to prevent the socket contacts being live when accessible.

The text of this standard is based on the following documents:

FDIS	Report on voting
64/2162/FDIS	64/2186/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

The reader's attention is drawn to the fact that Annex A lists all of the "in-some-country" clauses on differing practices of a less permanent nature relating to the subject of this standard.

A list of all parts in the IEC 60364 series, published under the general title *Low-voltage electrical installations*, can be found on the IEC website.

Future standards in this series will carry the new general title as cited above. Titles of existing standards in this series will be updated at the time of the next edition.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

INTRODUCTION

For the purpose of this part (IEC 60364-7-708) the requirements of the general Parts 1 to 6 of IEC 60364 apply.

The IEC 60364-7-7XX parts of IEC 60364 contain particular requirements for special installations or locations which are based on the requirements of the general parts of IEC 60364 (IEC 60364-1 to IEC 60364-6). These IEC 60364-7-7XX parts are considered in conjunction with the requirements of the general parts.

The particular requirements of this part of IEC 60364 supplement, modify or replace certain of the requirements of the general parts of IEC 60364 being valid at the time of publication of this part. The absence of reference to the exclusion of a part or a clause of a general part means that the corresponding clauses of the general part are applicable (undated reference).

Requirements of other 7XX parts being relevant for installations covered by this part also apply. This part may therefore also supplement, modify or replace certain of these requirements valid at the time of publication of this part.

The clause numbering of this part follows the pattern and corresponding references of IEC 60364. The numbers following the particular number of this part are those of the corresponding parts, or clauses of the other parts of the IEC 60364 series, valid at the time of publication of this part, as indicated in the normative references of this document (dated reference).

If requirements or explanations additional to those of the other parts of the IEC 60364 series are needed, the numbering of such items appears as 708.101, 708.102, 708.103 etc.

In the case where new or amended general parts with modified numbering were published after this part was issued, the clause numbers referring to a general part in this 701 part may no longer align with the latest edition of the general part. Dated references should be observed.

LOW-VOLTAGE ELECTRICAL INSTALLATIONS –

Part 7-708: Requirements for special installations or locations – Caravan parks, camping parks and similar locations

708 Caravan parks, camping parks and similar locations

708.1 Scope

The particular requirements contained in this part of IEC 60364 apply only to circuits intended to supply leisure accommodation vehicles, tents or residential park homes in caravan parks, camping parks and similar locations.

NOTE For the purposes of this document caravan park includes camping parks and similar locations.

The particular requirements do not apply to the internal electrical installations of leisure accommodation vehicles, mobile or transportable units or residential park homes.

708.2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60038, *IEC standard voltages*

IEC 60309-1:1999, *Plugs, socket-outlets and couplers for industrial purposes – Part 1: General requirements*

IEC 60309-1:1999/AMD1:2005

IEC 60309-1:1999/AMD2:2012

IEC 60309-2, *Plugs, socket-outlets and couplers for industrial purposes – Part 2: Dimensional interchangeability requirements for pin and contact-tubes accessories*

IEC 60309-4:2006, *Plugs, socket-outlets and couplers for industrial purposes – Part 4: Switched socket-outlets and connectors with or without interlock*

IEC 60309-4:2006/AMD1:2012

IEC 60364-4-43, *Low-voltage electrical installations – Part 4-43: Protection for safety – Protection against overcurrent*

IEC 62262, *Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code)*

708.3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

708.3.1**leisure accommodation vehicle**

unit of living accommodation for temporary or seasonal occupation that may meet the requirements for the construction and use of road vehicles

708.3.1.1**caravan**

trailer leisure accommodation vehicle, used for touring, that meets the requirements for the construction and use of road vehicles

708.3.1.2**motor caravan**

self-propelled leisure accommodation vehicle, used for touring, that meets the requirements for the construction and use of road vehicles

Note 1 to entry: The motor caravan is either adapted from a series production vehicle, or designed and built on an existing chassis, with or without the driving cab, the accommodation being either fixed or demountable.

708.3.1.3**mobile home**

transportable leisure accommodation vehicle that includes means for mobility but does not meet the requirements for the construction and use of road vehicles

708.3.2**caravan/tent pitch**

plot of ground intended to be occupied by a leisure accommodation vehicle or tent

708.3.3**caravan park****camping park**

area of land that contains two or more caravan or tent pitches.

708.3.4**residential park home**

factory-produced relocatable dwelling

708.31 Purposes, supplies and structure**708.312 Conductor arrangement and system earthing****708.312.2 Types of system earthing****708.312.2.1 TN-systems**

Add the following:

For a TN-system, the final circuit for the supply to a leisure accommodation vehicle, tent or residential park home shall not include a PEN conductor.

708.313 Supplies**708.313.1 General****708.313.1.101**

The nominal supply system voltage shall be selected from IEC 60038.

The nominal supply voltage of the installation for the supply of leisure accommodation vehicles shall not exceed 230 V AC single-phase and/or 400 V AC three-phase and 48 V DC.

708.4 Protection for safety

708.41 Protection against electric shock

708.410.3 General requirements

708.410.3.5

Add the following:

The protective measures of obstacles and placing out of reach as specified in Annex B of IEC 60364-4-41:2005 shall not be used.

708.410.3.6

Add the following:

The protective measures of non-conducting location and earth-free local equipotential bonding as specified in Annex C of IEC 60364-4-41:2005 shall not be used.

708.415 Additional protection

708.415.1 Residual current protective devices (RCDs)

Add the following:

Every socket-outlet shall be individually protected by an RCD having a rated residual operating current not exceeding 30 mA. Devices selected shall disconnect all live conductors.

A final circuit intended for the fixed connection for a supply to a mobile home or a residential park home shall be individually protected by an RCD having a rated residual operating current not exceeding 30 mA. Devices selected shall disconnect all live conductors.

708.5 Selection and erection of electrical equipment

NOTE See IEC TS 61439-7.

708.51 Selection and erection of electrical equipment – Common rules

708.512 Operational conditions and external influences

708.512.2 External influences

Add the following:

NOTE In a caravan park or camping park, special consideration is given to the protection of people, due to the fact that the human body can be in contact with earth potential, to the protection of wiring due to tent pegs or ground anchors and to the movement of heavy or high vehicles.

708.512.2.1.101 Presence of water (AD)

Equipment shall be selected with a degree of protection of at least IPX4 in order to protect against water splashes (AD4).

708.512.2.1.102 Presence of solid foreign bodies (AE)

Equipment shall be selected or provided with a degree of protection of at least IP4X in order to protect against the ingress of very small objects (AE3).

708.512.2.1.103 Impact (AG)

Equipment installed in a campsite shall be protected against mechanical damage (impact of high severity AG3). Protection of the equipment shall be afforded by one or more of the following:

- the position or location shall be selected to avoid damage by any reasonably foreseeable impact;
- local or general mechanical protection shall be provided;
- equipment that complies with a minimum degree of protection against external mechanical impact of IK08 (see IEC 62262) shall be installed.

708.52 Wiring systems**708.521 Types of wiring systems****708.521.7 Several circuits in one cable****708.521.7.101 Supply for the caravan or tent pitch**

The preferred method of supply for feeding the caravan pitch or tent pitch electrical supply equipment is by means of underground distribution circuits.

708.521.7.102 Underground cables

An underground distribution circuit shall, unless provided with additional mechanical protection, be buried at a sufficient depth to avoid being damaged, for example by tent pegs or ground anchors or by the movement of vehicles.

A depth of 0,6 m is generally considered as a minimum depth to fulfill this requirement. Alternatively the cable may be installed outside the pitch or other area where tent pegs or ground anchors may be driven.

NOTE For conduit systems buried underground, see IEC 61386-24.

708.521.7.103 Overhead cables or overhead insulated conductors

Every overhead conductor shall be insulated.

Poles and other supports for overhead wiring shall be located or protected so that they are unlikely to be damaged by any foreseeable movement of vehicles.

Every overhead conductor shall be at a height above ground of not less than 6 m in all areas subject to the movement of vehicles and 3,5 m in all other areas.

708.53 Isolation, switching and control**708.533 Devices for protection against overcurrent**

In addition the following applies:

Every socket-outlet shall be individually protected by an overcurrent protective device, in accordance with the requirements of IEC 60364-4-43.

A fixed connection for a supply to a mobile home or residential park home shall be individually protected by an overcurrent protective device, in accordance with the requirements of IEC 60364-4-43.

708.536 Isolation and switching**708.536.2 Isolation****708.536.2.1 General****708.536.2.1.1**

Add the following:

At least one means of isolation shall be installed in each distribution cabinet. This device shall disconnect all live conductors.

708.55 Other equipment**708.55.101 Socket-outlets**

708.55.101.1 Every socket-outlet shall comply with IEC 60309-2.

Every socket-outlet shall meet the degree of protection of at least IP44 or such protection shall be provided by an enclosure.

To prevent the socket contacts being live when accessible, every socket-outlet or connector shall either comply with IEC 60309-2 and shall be interlocked and classified according to IEC 60309-1:1999, 6.1.5 or shall be part of an interlocked self-contained product complying with IEC 60309-4 and classified according to IEC 60309-4:2006, 6.1.101 and 6.1.102.

708.55.101.2 Every socket-outlet shall be located as close as practicable to the caravan pitch or tent pitch to be supplied.

Socket-outlets shall be installed in the distribution board or in separate enclosures.

708.55.101.3 In order to avoid any hazard due to long connection cables no more than four socket-outlets shall be grouped together in any one enclosure. To ensure that the degree of protection is maintained when the socket-outlets are in use, no more than four socket-outlets shall be mounted within the same enclosure.

Columns (pillars) should be located so as to minimize the need for extension cables to cross gangways, etc.

708.55.101.4 Every caravan pitch or tent pitch shall be supplied with at least one socket-outlet.

708.55.101.5 The current rating of socket-outlets shall be not less than 16 A.

708.55.101.6 The lowest part of any socket-outlet shall be placed at a height between 0,5 m and 1,5 m from the ground. In special cases of extreme environmental conditions, it is permitted to exceed the stated maximum height of 1,5 m. In such cases, special measures shall be taken to ensure the safe insertion and removal of plugs.

NOTE This can be necessary if the caravan park or camping park risks being flooded. This can also be necessary if the caravan park is used during winter after heavy snow falls.

Annex A (informative)

List of notes concerning certain countries

Clause/Subclause	Text
708.41	Add the following notes at the end of the subclause: In the United Kingdom, UK law requires that the distributor shall not connect his combined neutral and protective conductor to any metalwork in a caravan or boat (Regulation 9(4) of The Electricity Safety, Quality and Continuity Regulations 2002 No. 2665).
708.512.2.1.103	Add the following note at the end of the subclause: NOTE In the United States, where provided on back-in sites, the caravan vehicle site electrical supply equipment shall be located on the left (road or driver) side of the parked vehicle, on a line that is 1,5 m to 2,1 m (5 ft to 7 ft) from the left edge (driver's side of the parked RV) of the stand and shall be located at any point on this line from the rear of the stand to 4,5 m (15 ft) forward of the rear of the stand. For pull-through sites, the electrical supply equipment shall be permitted to be located at any point along the line that is 1,5 m to 2,1 m (5 ft to 7 ft) from the left edge (driver's side of the parked RV) from 4,9 m (16 ft) forward of the rear of the stand to the centre point between the two access roads. The left edge (driver's side of the parked RV) of the stand shall be marked. All site supply equipment shall be accessible by an unobstructed entrance or passageway not less than 600 mm (2 ft) wide and 2,0 m (6 ft 6 in.) high.
708.521.7.102	Add the following notes at the end of the subclause: NOTE 2 In the Czech Republic and Slovakia the minimum required depth of burying non-mechanically protected cables is 0,7 m (with a warning strip 0,2 m to 0,3 m above the cable). NOTE 3 In Finland and Iceland, underground cables shall be buried at a depth of at least 0,7 m, unless such cables have additional mechanical protection, in which case the depth shall be at least 0,5 m. NOTE 4 In Japan and Switzerland a depth of 0,6 m is considered as a minimum depth to fulfill this requirement. NOTE 5 In Canada the minimum required depth of burying non-mechanically protected cables is 0,6 m (with a warning stripe at approximately 0,15 m above the cable). NOTE 6 In Ireland, a minimum depth of 500 mm is required, and 750mm where there is movement of heavy vehicles.
708.521.7.103	Add the following notes at the end of the subclause: NOTE 1 In Sweden, the minimum height for overhead conductors is 4,5 m. NOTE 2 In Finland and Japan, overhead conductors shall be at a height above ground of not less than 4 m in all other areas. NOTE 3 In Iceland, every overhead conductor shall be at a height above the ground of not less than 6 m in all areas subject to movement of vehicles and 4,5 m in all other areas. NOTE 4 In Ireland, a minimum height of 6 m is required throughout a caravan park. NOTE 5 In Ireland, they (conductors) shall be located at a minimum distance of 2 m outside the boundary of any caravan pitch.
708.536.2.1.1	Add the following note at the end of the subclause: NOTE In the United States, one isolating switching device shall be installed in each distribution cabinet. This switching device shall disconnect all live conductors, entering the cabinet, including the neutral conductor. One isolating switching device shall be installed for a maximum of every four socket-outlets.
708.55.101.1	Add the following note after the first paragraph: NOTE In Canada and the United States, socket-outlets rated 120 V or 120 V to 260 V according to other standards are used.

Bibliography

IEC 60364-7-721, *Low voltage electrical installations – Part 7-721: Requirements for special installations or locations – Electrical installations in caravans and motor caravans*

IEC 61386-24, *Conduit systems for cable management – Part 24: Particular requirements – Conduit systems buried underground*

IEC TS 61439-7, *Low-voltage switchgear and controlgear assemblies – Part 7: Assemblies for specific applications such as marinas, camping sites, market squares, electric vehicles charging stations*

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